



SRI VENKATESWARA COLLEGE OF ENGINEERING

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

in association with

ELECTRONICS AND COMMUNICATION ENGINEERS ASSOCIATION

along with

INSTITUTE OF ELETRONICS AND TELECOMMUNICATION ENGINEERS-STUDENTS' FORUM

&

RESEARCH ASSOCIATION FOR INNOVATIVE DESIGN IN COMMUNICATION AND ELECTRONICS

UPAGRAHA'26

SORCERER REALM (PAPER PRESENTATION)

TEAM NAME	FARMSPARK
PAPER TITLE	AI BASED REAL-TIME CROP RISK PREDICTION SYSTEM
PAPER DOMAIN	Machine Learning
TEAM MEMBERS	VISHAL .P, SETHU MADHAVAN S.B , NARENDRAN .V

TITLE: AI BASED REAL-TIME CROP RISK PREDICTION SYSTEM

OBJECTIVE:

To create an intelligent, web-based system predicting crop risk levels in real time using machine learning. Which integrates real-time weather data, soil parameters, stock availability, and government schemes. Provides cultivation guidance for beginners, live updates, chatbot as helpdesk and multilingual support (English & Tamil).

METHODOLOGY:

Our methodology includes collecting soil and weather datasets, preprocessing and integrating them, selecting agronomic features, training a crop recommendation model, deploying it using FastAPI, building a multilingual web interface, and integrating a RAG-based chatbot for agricultural guidance. Real-time weather APIs and district soil mappings are used to auto-fill inputs and improve prediction accuracy & also providing a graphical representation which represents the probability of chosen crop by the farmer and recommended crop from the model by using bar graphs/pie chart for visualization. Additionally, we support farmers to have a right knowledge in schemes for them in India and helping them to buy products like seeds and agricultural equipments at the right time and right place in our website.

EXPECTED OUTCOME:

The proposed system is expected to provide an intelligent, farmer-friendly web platform that delivers accurate crop recommendations based on integrated soil nutrient data and real-time weather parameters. The trained machine learning model will generate suitable crop suggestions with improved prediction accuracy through automated district and area-based data auto-filling.

The platform will present prediction results along with graphical visualizations such as bar charts and pie charts to clearly show the suitability probability of the farmer's selected crop versus the model-recommended crops. A multilingual interface will enable accessibility for regional language users.

An integrated RAG-based chatbot will provide reliable agricultural guidance, crop information, and awareness about Indian government schemes. The system will also support farmers by helping them identify the right time and place to purchase seeds and agricultural equipment, thereby improving decision-making and agricultural productivity.

Conclusion:

- Farmers gain timely insights into crop suitability, weather conditions, and soil parameters through data-driven recommendations.
- The system improves decision-making for crop cultivation, input planning, stock management, and effective utilization of government schemes.
- A user-friendly, multilingual interface makes the platform accessible to both beginner and experienced farmers.
- Visual analytics and intelligent assistance reduce uncertainty and stress caused by unpredictable environmental conditions.
- The integrated recommendation model and knowledge chatbot together provide continuous decision support for better agricultural outcomes.

